問題4

$\dot{I} = 10\exp(j\frac{\pi}{6})$

$\dot{V}_R = R\dot{I} = 50\exp(j\frac{\pi}{6})$

$v_R = 50\sqrt{2}\sin(100t + \frac{\pi}{6})$

$\dot{V}_L = X_L\dot{I} = j2\pi f \cdot 50\times10^{-3} \times 10\exp(j\frac{\pi}{6})$

$= 0.1\pi f \exp(j\frac{\pi}{2}) \times 10\exp(j\frac{\pi}{6})$

$= \pi f \exp(j\frac{4}{6}\pi) = \frac{\pi f}{100\pi}\exp(j\frac{2}{3}\pi)$

$v_R = \frac{\pi f}{100\pi}\sin(100t + \frac{2}{3}\pi)$

$\dot{V}_C = X_C\dot{I} = \frac{1}{j2\pi f \cdot 2\times10^{-7}} \times 10\exp(j\frac{\pi}{6})$

$= -j\frac{1}{2\pi\times100\times2\times10^{-7}} \times 10\exp(j\frac{\pi}{6})$

$= \frac{1}{4\pi}\times\exp(-j\frac{\pi}{2}) \times 10\exp(j\frac{\pi}{6})$

$= \frac{1}{4\pi}\exp(-j\frac{\pi}{3})$

$v_C = \frac{1}{4\pi}\sin(100t - \frac{\pi}{3})$

$16\sqrt{2}\exp(-j\frac{\pi}{4})$

問図 5-39